

Q-1: K^{th} Largest Element at Every Step

You are given an integer array *nums* of size N and an integer K . Consider the array elements one by one from left to right. Find the K^{th} largest element among the elements from the **beginning** up to (including) the current element. Repeat this till the end of array.

If K^{th} largest element cannot be found, represent it by -1.

The results are stored and returned in a vector of size N .

Constraints:

- $1 \leq K \leq N$

Expected Time Complexity: $\Theta(N \log(N))$

Example Test Cases:

TEST CASE 1

Input: $N = 3$, $K = 3$

nums = [100,500,300]

Output: [-1, -1, 100]

Explanation:

Considering elements from left to right:

After elements [100], there is no 3rd largest element, hence -1.

After elements [100,500], there is no 3rd largest element, hence -1.

After elements [100,500,300], 3rd largest element is 100.

Thus, the final output is [-1,-1,100].

Input Format:

- The first line contains two integers N and K - the size of the array *nums* and the K^{th} largest element you need to find at each step.
- The second line contains N space-separated integers of the array *nums*.

Output Format: Return a vector of size N .

You are free to use any inbuilt function

Q-2: Smallest Range Covering Elements from K Lists

You have k lists of *sorted* integers in **non-decreasing** order. Find the **smallest** range that includes at least one number from each of the k lists.

Note: We define that the range $[a, b]$ is smaller than the range $[c, d]$ if $b - a < d - c$ **or** $a < c$ if $b - a == d - c$.

The k lists have been provided in a 2-D vector *nums*

Constraints:

- The size of the 2D vector *nums* is **equal** to k .
- *nums*[*i*] is sorted in **non-decreasing** order.

Example Test Cases:

TEST CASE 1

Input: *nums* = [[4,10,15,24,26], [0,9,12,20], [5,18,22,30]]

Output: [20, 24]

Explanation:

List 1: [4, 10, 15, 24,26], 24 is in range [20,24].

List 2: [0, 9, 12, 20], 20 is in range [20,24].

List 3: [5, 18, 22, 30], 22 is in range [20,24].

TEST CASE 2

Input: *nums* = [[1,2,3], [1,2,3], [1,2,3]]

Output: [1,1]

Expected Time Complexity: $\Theta(n \cdot \log k)$ where n is the total number of elements across all the lists.

Input Format:

1. The first line contains an integer k - the number of lists/ size of array *nums*.
2. The next k lines represent elements in the list these elements are sorted in non - decreasing order.

Output Format:

Your output must be a vector of size 2 indicating the required range.

`You are free to use any inbuilt function`

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)